7 (a) The electromotive force (e.m.f.) of a source is defined as the amount of energy transferred from non-electrical forms of energy to electrical energy per unit charge as the charge passes through a complete circuit. [B1]

The potential difference *V* between two points of a circuit is defined as the amount of electrical energy converted to **other** forms of energy **per unit charge** moved between these two points. [B1]

(i)

Total resistance, $R_T = 1.0 + \frac{1}{\frac{1}{2.0}+\frac{1}{6.0}} = 2.5\ \Omega$ [M1]

$E = I R_T$ [M1]

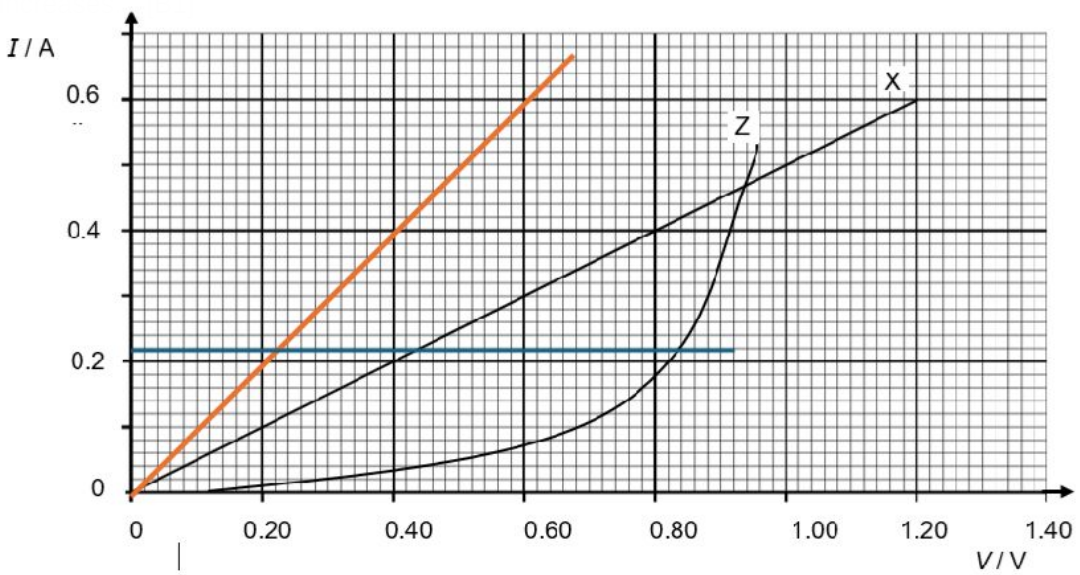
$I = 1.5 / 2.5 = 0.60$ A. (Shown) [A0]

(ii) Energy dissipated = $I^2 r t$ [M1]

$= (0.60)^2 (1.0)(8.0 \times 60) = 172.8 = 170$ J [B1]

(iii) When temperature increased and resistance of Z decreases, the total current in the cell will increase. Thus, the power dissipated in the cell

(iv)



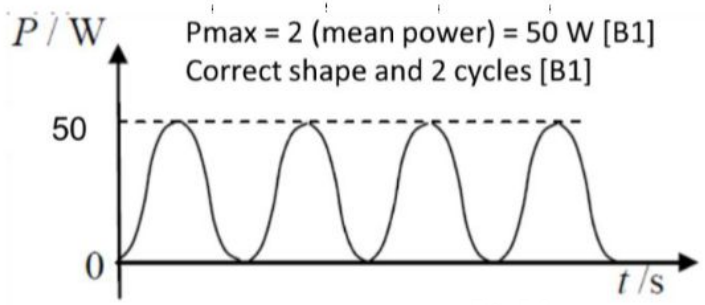
MARGIN

Since the components are connected in series, the current passing through the components are the same, draw a horizontal line to find the total p.d. across equal to 1.5 V. [B1]

p.d across the X and 1.0 Ω is 0.66V, and p.d across Z is 0.84V [B1]

v Thermistor [B1]

7 (c)i



Pmax = 2 (mean power) = 50 W [B1]
Correct shape and 2 cycles [B1]

cii Sketch of P-t graph with Illustration of area of rectangle of height $1/2P_o$.
B1
Explain that the area of the rectangle is the same as the area of the P-t curve
B1

(d) (i) $\frac{Ns}{Np} = \frac{Vs}{Vp}$

Vs = 1/45 (240) = 5.33V

Hence $V_o = \sqrt{2} \times 5.33 = 7.54$ V [B1]

Assumption : Transformer is ideal / 100% efficiency [B1]

(ii) The voltage step down to the secondary coil is sinusoidal but with the diode in place, the voltage across the battery is half wave rectified . This imply that the time that the battery get charged is only half of the time voltage is step down

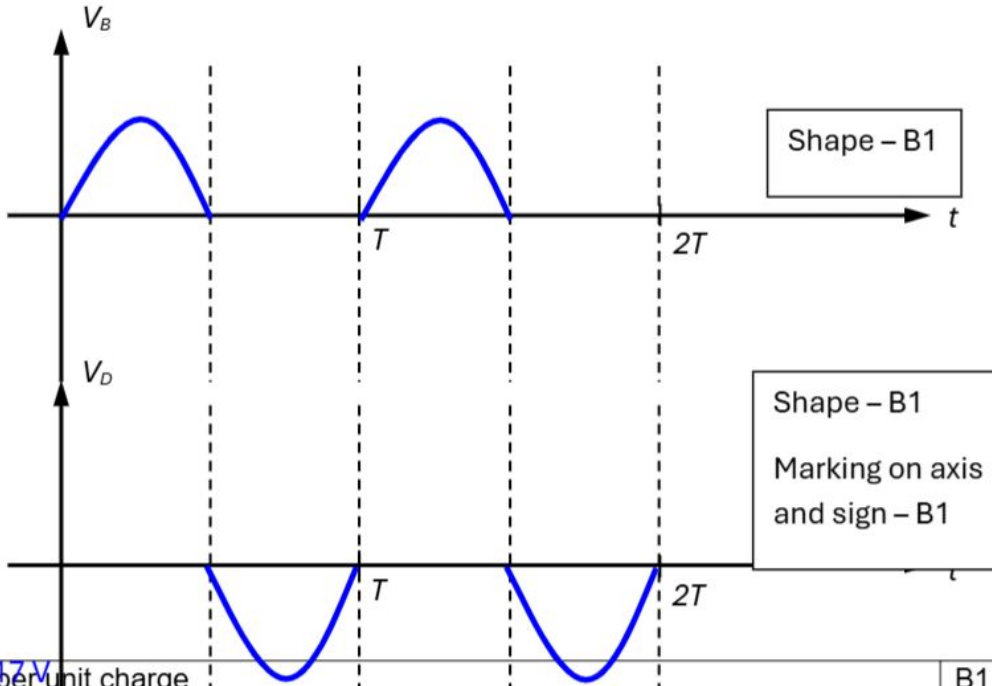
(iii)

Without diode , $P = (5.3)^2/R$

With diode , power delivered = $P/2 = V'^2_{rms}/R$, $28.1/2R = V'^2_{rms}/R$ [B1]

Hence V'_{rms} $\sqrt{(28.1/2)} = 3.74$ V

(iv)



1]

(v)

8a	force per unit charge	B1